

Curriculum Vitae

Ph.D. Candidate Lien Cartaya Bermúdez

Institute of Mathematics, University of Talca, Chile

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Research Interests

Representation theory, reflection groups, non-commutative algebra, path algebras, combinatorics.

Publications

- L. Cartaya, S. Griffeth. *Zero fibers of quaternionic quotient singularities*. Rev. Mat. Iberoam. 41 (2025), no. 3, pp.1461-1487, DOI: 10.4171/RMI/1528.

Education

2024–2028^(exp) **Ph.D. in Mathematics**, Universidad de Chile, Chile.

Advisor: Stephen Griffeth (Representation Theory). Ph.D. Candidate since Jan 2025.

2022–2024 **M.Sc. in Mathematics**, Universidad de Talca, Chile.

Thesis: *Non-commutative deformations of quotient singularities* (review of an article). Advisor: Stephen Griffeth.

2016–2020 **Degree in Mathematics**, Universidad de La Habana, Cuba.

Thesis: *Study on high-dimensional algorithms for the reduction of STRESS in Multidimensional Scaling*. Advisor: Elina Miret Barroso.

Awards and Scholarships

- 2026: Operational expenses for research, ANID, Chile.
- 2024–2028: ANID Beca de Doctorado Nacional, Chile.

Appointments

- 2023–2025 Thesis student, FONDECYT Regular Project 1231355, Chile.

Talks and Posters

Oct 2025 Oral: *Walking along quivers towards finitely generated algebras.*, Seminario de Estudiantes de la Universidad de Chile (Seminario Relajao), Santiago de Chile, Chile. Seminario de Estudiantes de la Universidad de Valparaíso (Puerto de Matemáticas), Valparaíso, Chile.

Aug 2025 Poster: *The diagonal coinvariant ring for quaternionic reflection groups*, X Encuentro Nacional de Álgebra – EMALCA Argentina, Córdoba.

Apr 2025 Oral: *About the diagonal coinvariant ring for quaternionic reflection groups*, XXXVII Jornada de Matemática de la Zona Sur. Universidad Católica del Maule, Talca, Chile.

Apr 2025 Oral: *Understanding reflection groups from their representations*, Seminario No-Degenerado. Instituto de Matemáticas, Universidad de Talca, Chile.

- Jul 2024 Oral: *Zero fiber of quotient singularities*, XXIV Coloquio Latinoamericano de Álgebra (CLA), Santiago de Chile.
- May 2024 Oral: *Quivers: a preliminary vision*, Seminario No-Degenerado. Instituto de Matemáticas, Universidad de Talca, Chile.
- Mar 2024 Oral: *Zero fiber of quotient singularities*, 2do Encuentro Nacional de Estudiantes de Matemáticas. (A National Meeting of Graduate Mathematics Students. Instituto de Matemáticas, Universidad de Talca, Chile.
- Aug 2023 Oral: *McKay correspondence*, Seminario No-Degenerado. Instituto de Matemáticas, Universidad de Talca, Chile.
- Jul 2021 Poster: *Application of multidimensional scaling to SARS-CoV-2 patients treated with Interferon*, I Festival Nacional de Matemática y Computación, La Habana, Cuba.
- Mar 2020 Oral: *Study and implementation of algorithms for high-dimensional multidimensional scaling*, IX Encuentro de Matemática Numérica y Optimización, Cuba-México, La Habana, Cuba.

Teaching Experience

- March 2026 Preparatory Mathematics course for engineering degrees, Universidad de Talca. (duration 1 week)
- 2025 Lecturing Experience, Universidad de Talca. Precalculus, Calculus (Commercial Engineering).
- 2022–2025 Teaching Assistant, Universidad de Talca. Statistics & Probability, Calculus, Linear Algebra, ODEs.
- 2017–2022 Teaching Assistant, Universidad de La Habana. Calculus, Linear Algebra, Real Analysis.
- Jan 2023 Teaching Assistant, Summer School (Finite subgroups of SL_2 , by S. Griffeth), Universidad de Talca.

Academic Service & Outreach

- Jan 2026 Instructor of the short course *Invisible Patterns: Changing without changing* during the Campamento Hannah. Instituto de Matemáticas, Universidad de Talca, Chile. (mathematics outreach program for female high school students)
- Jan 2026 Member of the Organizing Committee of the 2026 Summer School for Undergraduates in Mathematics. Instituto de Matemáticas, Universidad de Talca, Chile. (Academic planning and coordination for undergraduate participants.)

Schools

- Mar 2023 CIMPA School *Crossroads of Geometry, Representation Theory and Higher Structures*, Puerto Madryn, Argentina.

Additional Experience

- 2024–2026 Social Media Assistant (part-time), Institute of Mathematics, University of Talca.
- 2021 Data Scientist (part-time), Mandao, La Habana, Cuba.

References

- Stephen Griffeth, Professor of Mathematics, Universidad de Talca, Chile. Email: sgriffeth@utalca.cl
- María Ofelia Ronco, Universidad de Talca, Chile. Email: mronco@utalca.cl

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